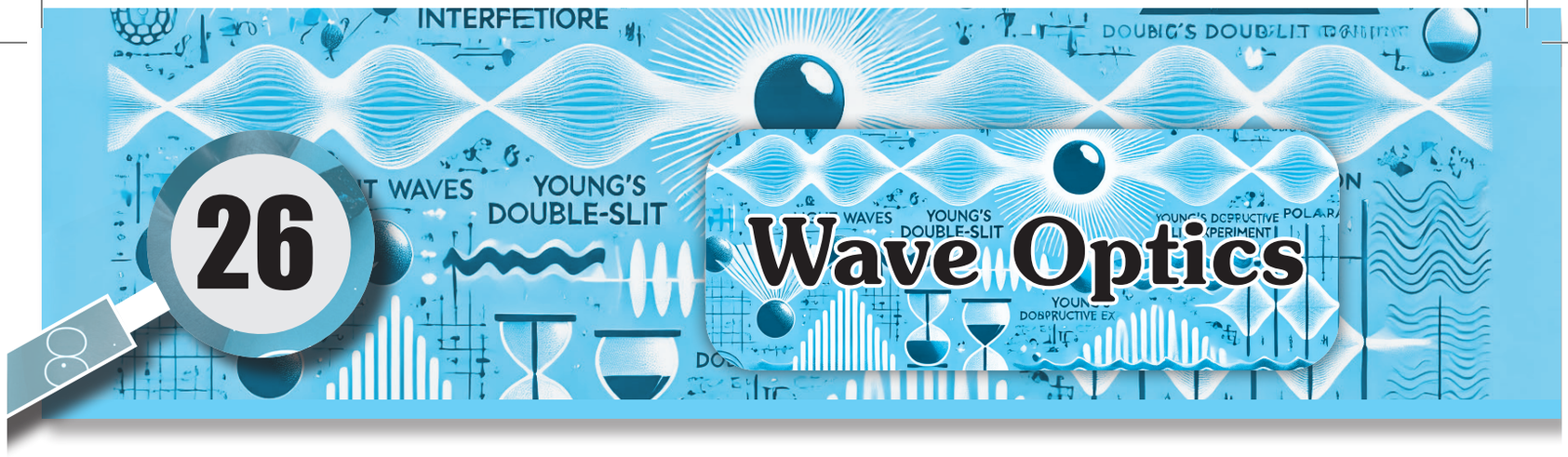


Huygen's Wave Theory

1. How light travel from sun to earth if there is vacuum between them?

Wave Front

2. For light diverging from point source :

(i) Wavefront is spherical

(ii) Intensity decreases in proportion to distance squared.

(iii) Wavefront is parabolic.

(iv) Intensity at wavefront doesn't depend on distance.

(*a*) (i), (ii) (*b*) (i), (iii)

(*c*) (ii), (iii) (*d*) (iii), (iv)

3. A wavefront of a wave has direction with wave motion

(*a*) Parallel (*b*) Perpendicular

(*c*) At 45° (*d*) Opposite

4. Wavefront is a locus of all the point vibrate with same:

(*a*) Amplitude (*b*) Phase

(*c*) Period (*d*) Frequency

5. A wavefront is parallel to line $Y = -x + 4$ then direction of propagation wave with $+x$ axis is:

(*a*) 120° (*b*) 45° (*c*) 145° (*d*) 125°

6. The wavefront of source which is at infinity is:

(*a*) Spherical (*b*) Cylindrical

(*c*) Planer (*d*) Circular

7. Light emerges out of a convex lens when a source of light kept at its focus. The shape of wavefront of the light is:

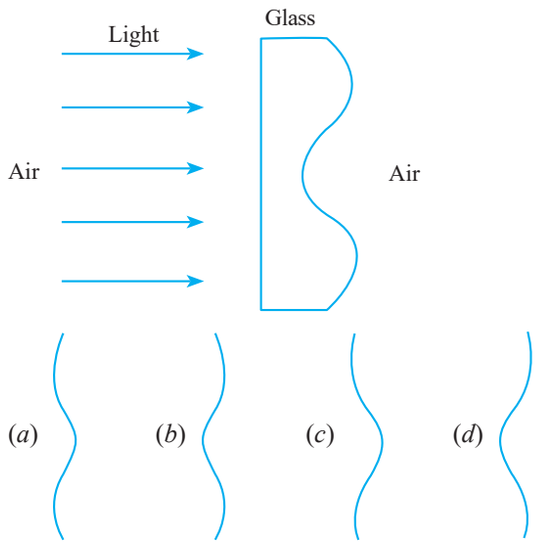
(*a*) Both spherical and cylindrical

(*b*) Cylindrical

(*c*) Spherical

(*d*) Plane

8. A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wave front will be (figures are schematic and not drawn to scale)



9. Normal to the wave front represents a ray of light.

1. A wave front always travels in forward in a medium
2. Phase difference between various particles on a wave front is zero:
3. A point source of light gives rise to spherical wave fronts.

Correct statements are:-

(*a*) 1 and 2 only (*b*) 1 and 3 only

(*c*) 3 and 2 only (*d*) All are correct

Superposition of Two Waves

10. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the direction. The sum of the maximum and minimum intensities is:

(*a*) $I_1 + I_2$ (*b*) $\left(\sqrt{I_1} + \sqrt{I_2}\right)$

(*c*) $\left(\sqrt{I_1} + \sqrt{I_2}\right)^2$ (*d*) $2(I_1 + I_2)$

11. The path difference between two waves $y_1 = A_1 \sin \omega t$ and $y_2 = A_2 \cos(\omega t + \phi)$ will be

(*a*) $(\lambda/2\pi)\,\phi$ (*b*) $(\lambda/2\pi)(\phi + \pi/2)$

(*c*) $(2\pi\lambda)(\phi - \pi/2)$ (*d*) $(2\pi/\lambda)$

12. The displacements of two interfering light waves are $y_1 = 4 \sin(\omega t)$ and $y_2 = 3 \cos(\omega t)$. The amplitude of the resultant wave is (y_1 and y_2 are in CGS system).

(*a*) 5 cm (*b*) 7 cm (*c*) 1 cm (*d*) Zero

13. Two sources of intensity of 2I and 8I are used in an interference experiment. The intensity at a point where the waves from two sources superimpose with a phase difference of (*i*) zero (*ii*) $\pi/2$ and (*iii*) π is:

(*a*) 18I, 10I, 2I (*b*) 5I, 4I, I

(*c*) 2I, I, I/2 (*d*) 2I, 10I, 18I

14. The intensity of interference waves in an interference pattern is same as i_0. The resultant intensity at the point of observation will be:

(*a*) $I = 2I_0 [1 + \cos \phi]$ (*b*) $I = I_0 [1 + \cos \phi]$

(*c*) $I = \dfrac{[1 + \cos \phi]}{I_0}$ (*d*) $\dfrac{[1 + \cos \phi]}{2I_0}$

15. The resultant amplitude in interference with two coherent source depends upon:

(*a*) Intensity (*b*) Only phase difference

(*c*) On both the above (*d*) None of the above

16. The necessary condition for phenomenon of interference to occur is:

(*a*) There should be two coherent sources.

(*b*) The frequency and amplitude of both the waves should be same.

(*c*) The propagation of waves should be simultaneously and in same direction.

(*d*) All of the above.

17. The equation for two waves obtained by two light sources are as given below:

$y_1 = A_1 \sin 3\omega t$, $y_2 = A_2 \cos(3\omega t + \pi/6)$. What will be the value of phase difference at the time t :

(*a*) $3\pi/2$ (*b*) $2\pi/3$ (*c*) π (*d*) $\pi/2$

18. Two coherent sources have intensity ratio of 100 : 1, and are used for obtaining the phenomenon of interference. Then the ratio of maximum and minimum intensity will be:

(*a*) 100 : 1 (*b*) 121 : 81 (*c*) 1 : 1 (*d*) 5 : 1

19. Two independent monochromatic sodium lamps cannot produce interference because:

(*a*) The frequencies of two sources are different.

(*b*) The phase difference between two sources changes with respect to time.

(*c*) The two sources become coherent.

(*d*) The amplitudes of two sources are different.

20. If two waves represented by $y_1 = 4 \sin \omega t$ and $y_2 = 3 \sin(\omega t + \pi/3)$ interference at a point, the amplitude of the resulting wave will be about:

(*a*) 7 (*b*) 6 (*c*) 5 (*d*) 3.5

21. If an interference pattern has maximum and minimum intensities in 36 : 1 ratio then what will be the ratio of amplitudes?

(*a*) 5 : 7 (*b*) 7 : 4 (*c*) 4 : 7 (*d*) 7 : 5

22. In Young's double slit interference experiment, using two coherent waves of different amplitudes, the intensities ratio between bright and dark fringes is 3. Then, the value of the ratio of the amplitudes of the wave that arrive there is:

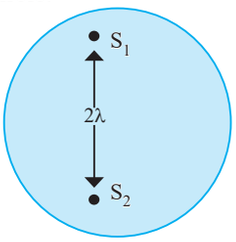
(*a*) $\left(\dfrac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ (*b*) $\left(\dfrac{\sqrt{3}-1}{\sqrt{3}+1}\right)$

(*c*) $\sqrt{3}$: 1 (*d*) 1 : $\sqrt{3}$

23. Two coherent source of intensity ratio α interfere. The value of $\dfrac{I_{max} - I_{min}}{I_{max} + I_{min}}$ is:

(*a*) $2\sqrt{\dfrac{\alpha}{1+\alpha}}$ (*b*) $\dfrac{2\sqrt{\alpha}}{1+\alpha}$ (*c*) $\dfrac{1+\alpha}{2\sqrt{\alpha}}$ (*d*) $\dfrac{1-\alpha}{1+\alpha}$

24. λ is wavelength of light, then find total number of maxima & minima on circle.



25. Two wave of equation $y_1 = 4 \sin(Kx + \omega t + 30°)$, $y_2 = 3 \sin(Kx + \omega t + 120°)$ then find resultant superimpose wave equation.

26. Two wave of equation $y_1 = 3 \sin(\omega t + Kx)$

$y_2 = 3 \cos(Kx + \omega t + 30°)$

27. $y_1 = 5 \sin(Kx + \omega t - 30°)$, $y_2 = 5 \cos(Kx + \omega t)$

Find $y = y_1 + y_2$.

28. $y_1 = 4 \sin(Kx + \omega t + 30°)$, $y_2 = 4 \cos(Kx + \omega t)$

Find $y = y_1 + y_2$.

29. In Fresnel's biprism experiment the width of 10 fringes is 2 cm which are formed at a distance of two meter from the slit. If the wavelength of light is 5100 Å then the distance between two coherent sources.

30. Two incoherent sources of light emitting light of intensity I_0 and $3I_0$ interfere in a medium. Then the resultant intensity at any point will be:

31. Interference was observed in interference chamber where air was present, now the chamber is evacuated, and if the same light is used, a careful observer will see:

(*a*) no interference

(*b*) interference with brighter bands

(*c*) interference with dark bands

(*d*) interference fringe with larger width

32. Two coherent monochromatic light beams of intensities I and 4I are superimposed. The difference between maximum and minimum possible intensities in the resulting beam is xI. The value of x is_____

33. To demonstrate the phenomena of interference, we require:
(*a*) two sources which emit radiation of same frequency.
(*b*) two sources which emit radiation of nearly same frequency
(*c*) two sources which emit radiation of the same frequency and have a definite phase relationship.
(*d*) two sources which emit radiation of different wavelenths.

Young's Double Slit Experiment (YDSE)

34. The wavelength of the light used in Young's double slit experiment is λ. The intensity at a point on the screen is I, where the path difference is λ/6. If I_0 denotes the maximum intensity, then the ratio of I and I_0 is:
(*a*) 0.866 (*b*) 0.5
(*c*) 0.707 (*d*) 0.75

35. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes:
(*a*) half (*b*) four times (*c*) one-fourth (*d*) double

36. In a Young's double slit experiment, if there is no initial phase difference between the light from the two slits, a point on the screen corresponding to the fifth minimum has path difference.
(*a*) $11\frac{\lambda}{2}$ (*b*) $5\frac{\lambda}{2}$ (*c*) $10\frac{\lambda}{2}$ (*d*) $9\frac{\lambda}{2}$

37. Two slits in Young's experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is:
(*a*) $\frac{121}{49}$ (*b*) $\frac{49}{121}$ (*c*) $\frac{4}{9}$ (*d*) $\frac{9}{4}$

38. In the Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is K, (λ being the wave length of light used). The intensity at a point where the path difference is λ/4, will be:
(*a*) K (*b*) $K/4$ (*c*) $K/2$ (*d*) Zero

39. The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10\ d$?
(*a*) I_0 (*b*) $I_0/9$
(*c*) $\frac{3}{4}I_0$ (*d*) $I_0/2$

40. In Young's double slit experiment carried out with light of wavelength (λ) = 5000 Å, the distance between the slits is 0.2 mm and the screen is at 200 cm from the slits. The central maximum is at $x = 0$. The third maximum (taking the central maximum as zeroth maximum) will be at x equal to:
(*a*) 1.67 cm (*b*) 1.5 cm
(*c*) 0.5 cm (*d*) 5.0 cm

41. In Young's experiment, two coherent sources are placed 0.90 mm apart and fringe are observed one metre away. If it produces second dark fringe at a distance of 1 mm from central fringe, the wavelength of monochromatic light used would be:
(*a*) 60×10^{-4} cm (*b*) 10×10^{-4} cm
(*c*) 10×10^{-5} cm (*d*) 6×10^{-5} cm

42. In a double slit experiment, when light of wavelength 400 nm was used the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? (μ = 4/3)
(*a*) 0.1° (*b*) 0.266°
(*c*) 0.15° (*d*) 0.05°

43. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to:
(*a*) 1.8 mm (*b*) 1.9 mm (*c*) 2.1 mm (*d*) 1.7 mm

44. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly:
(*a*) 1.59 (*b*) 1.69
(*c*) 1.78 (*d*) 1.25

45. In Young's double slit experiment, the slits are 2 mm apart and are illuminated by photons of two wavelengths $\lambda_1 = 12000$ Å and $\lambda_2 = 10000$ Å. At what minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other?
(*a*) 4 mm (*b*) 3 mm
(*c*) 8 mm (*d*) 6 mm

46. In a Young's double slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$ without disturbing the geometrical arrangement, the new fringe width will be:
(*a*) 0.30 mm (*b*) 0.40 mm
(*c*) 0.53 mm (*d*) 450 μm

47. The Young's double slit experiment is performed with blue and with green light of wavelengths 4360 Å and 5460 Å respectively. If x is the distance of 4th maxima from the central one, then:

(a) x (blue) = x (green) (b) x (blue) > x (green)

(c) x (blue) < x (green) (d) $\frac{x(\text{blue})}{x(\text{green})} = \frac{5460}{4360}$

48. Whose fringe width will be more, the one for red light or the one for yellow light, all other things be the same?

49. In a Young's double slit setup using monochromatic light of wavelength λ, the intensity of light at a point where path difference is 2λ is found to be I_0. What will be the intensity at a point where path difference is λ/3?

50. In YDSE, d = 2 mm, D = 2 m and λ = 500 nm. If intensity of two slits are I_0 and $9I_0$, then find intensity at $y = \frac{1}{6}$ mm.

(a) $7I_0$ (b) $10I_0$

(c) $16I_0$ (d) $4I_0$

51. In YDSE, Find the thickness of a glass slab (1.5 = μ) which should be placed as S_1 so the central maxima lies at a point where 5th bright fringes was lying earlier (λ = 5000 Å).

52. In YDSE, a thin film (μ = 1.6) of thickness 0.01 mm is placed at 1st slit, the central maxima shift to 10th bright fringes earlier, then wavelength of wave is?

53. Light of wavelength λ is incident on a slit of width d. The resulting diffraction pattern is observed on a screen at a distance D. The linear width of the central maxima equal to width of the slit if D equals.

(a) d/λ (b) $2\lambda/d$

(c) $d^2/2\lambda$ (d) $2\lambda^2/d$

54. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is:

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

55. Two wavelengths λ_1 and λ_2 are used in Young's double slit experiment λ_1 = 450 nm and λ_2 = 650 nm. The minimum order of fringe produced by λ_2 which overlaps with the fringe produced by λ_1 is n. The value of n is_____.

56. In a Young's double slit experiment, the intensities at two points, for the path difference $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0} = ?$

57. In young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate (μ = 1.5) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be:

(a) 3 (b) 2

(c) 1.5 (d) 0.5

58. If the source of light used in a Young's double slit experiment changed from red to violet:

(a) The fringes will become brighter.

(b) Consecutive fringe lines will come closer.

(c) The central bright fringe will become a dark fringe.

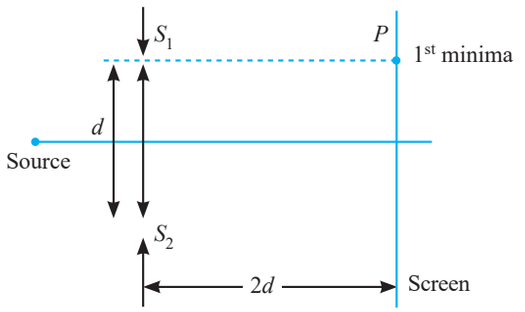
(d) The intensity of minima will increase.

59. In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of a wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be:

(a) 28 (b) 24

(c) 30 (d) 18

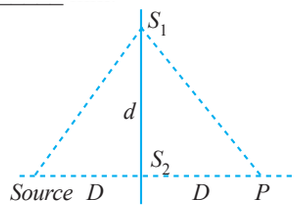
60. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)?



(a) $\frac{\lambda}{2(\sqrt{5}-2)}$ (b) $\frac{\lambda}{(\sqrt{5}-2)}$

(c) $\frac{\lambda}{2(5-\sqrt{2})}$ (d) $\frac{\lambda}{(5-\sqrt{2})}$

61. In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P. If D = 0.2 m the minimum distance between the slits S_1 and S_1 is _____ mm.



62. In YDSE, match the column-I and column-II.

Column-I		Column-II	
(i)	Fringe width	A.	$\frac{3\lambda D}{d}$
(ii)	Angular fringe width	B.	$\frac{7\lambda D}{2d}$
(iii)	Position of 3rd bright w.r.t. centre	C.	$\frac{\lambda}{d}$
(iv)	Position of 4th dark w.r.t. centre	D.	$\frac{\lambda D}{d}$

(*a*) i - A, ii - B, iii - C, iv - D
(*b*) i - D, ii - C, iii - A, iv - B
(*c*) i - D, ii - C, iii - B, iv - A
(*d*) i - C, ii - D, iii - A, iv - B

63. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): When width of one of the slits in Young's double slit experiment is nine times that of the other, then brighter fringes are four times brighter than the dark fringes.

Reason (R): The amplitude of the wave is proportional to the width of the slit.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is Not the correct explanation of Assertion (A).
(*c*) Assertion (A) is true but Reason (R) is false.
(*d*) Assertion (A) is false but Reason (R) is true.

64. The maximum number of possible interference maxima for slit separation equal to two and half times the wavelength in Young's double slit experiment in:

(*a*) Infinite (*b*) Five (*c*) Three (*d*) Zero

65. If Young's experiment is performed using two separate identical sources of light instead of using two slits and one light source then the:

(*a*) Interference fringes will be darker
(*b*) Interference fringes will be darker
(*c*) Fringes will not be obtained
(*d*) Contrast between bright and dark fringes increases.

66. In YDSE, if the source of light changes from orange to blue then

(*a*) Central bright fringe will become a dark fringe.
(*b*) Distance between consecutive fringes will decrease.
(*c*) Distance between consecutive fringe will increase.
(*d*) Intensity of the minima will increase

Diffraction

67. Angular width of the central maxima in the Fraunhoffer diffraction for $\lambda = 6000$ Å is θ_0. When the same slit is illuminated by another monochromatic light, the angular width decreases by 30%. The wavelength of this light is:

(*a*) 420 Å (*b*) 1800 Å (*c*) 4200 Å (*d*) 6000 Å

68. In a diffraction pattern due to a single slit of width '*a*', the first minimum is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of:

(*a*) $\sin^{-1}\left(\frac{1}{4}\right)$ (*b*) $\sin^{-1}\left(\frac{2}{3}\right)$
(*c*) $\sin^{-1}\left(\frac{1}{2}\right)$ (*d*) $\sin^{-1}\left(\frac{3}{4}\right)$

69. For a parallel beam of monochromatic light of wavelength 'λ' diffraction is produced by a single slit whose width '*a*' is of the wavelength of the light. If '*D*' is the distance of the screen from the slit, the width of the central maxima will be:

(*a*) $\frac{D\lambda}{a}$ (*b*) $\frac{Da}{\lambda}$ (*c*) $\frac{2Da}{\lambda}$ (*d*) $\frac{2D\lambda}{a}$

70. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is:

(*a*) $\pi/2$ radian (*b*) π radian
(*c*) $\pi/8$ radian (*d*) $\pi/4$ radian

71. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed in focal plane. The first minimum will be formed for the angle of diffraction equal to:

(*a*) 0° (*b*) 15° (*c*) 30° (*d*) 50°

72. A beam of light of $\lambda = 600$ nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is:

(*a*) 1.2 cm (*b*) 12 mm (*c*) 2.4 cm (*d*) 24 mm

73. A microwave of wavelength 2.0 cm falls normally on a slit of width 4.0 cm. The angular spread of the central maxima of the diffraction pattern obtained on a screen 1.5 m away from the slit, will be:

(*a*) 30° (*b*) 15° (*c*) 60° (*d*) 45°

74. A monochromatic light of wavelength 6000 Å is incident on the single slit of width 0.01 mm. If the diffraction pattern is formed at the focus of the convex lens of focal length 20 cm, the linear width of the central maximum is:

(*a*) 60 mm (*b*) 24 mm (*c*) 120 mm (*d*) 12 mm

75. Orange light of wavelength 6000×10^{-10}m illuminates a single slit of width 0.6×10^{-4} m. The maximum possible number of diffraction minima produced on both sides of the central maximum is:

76. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In case of single slit diffraction intensity of higher order maxima decreases.

Reason (R): Higher order maxima are at larger distance.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true

77. In a Fraunhoffer diffraction experiment at a single slit using light of wavelength 400 nm, the first minimum is formed at an angle of 30°. Then the direction 'θ' of the first secondary maximum is:

(*a*) $\tan^{-1}\left(\frac{4}{3}\right)$ (*b*) 60°

(*c*) $\sin^{-1}\left(\frac{3}{4}\right)$ (*d*) $\tan^{-1}\left(\frac{3}{4}\right)$

Polarisation

78. Two polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is:

(*a*) $I_0/2$ (*b*) $I_0/4$ (*c*) $I_0/8$ (*d*) $I_0/16$

79. Two nicols are oriented with their principal planes making an angle of 60°. The percentage of incident unpolarised light which passes through the system is:

(*a*) 50% (*b*) 100% (*c*) 12.5% (*d*) 37.5%

80. Two polaroids are oriented with their planes perpendicular to incident light and transmission axis making an angle of 30° with each other. What fraction of incident unpolarised light is transmitted?

(*a*) 1/2 (*b*) 3/4 (*c*) 7/8 (*d*) 3/8

81. Which of the following phenomena is not explain by wave nature of light.

A. reflection B. diffraction

C. photoelectric effect D. interference

E. polarization

(*a*) E only (*b*) C only (*c*) B, D only (*d*) A. C only

82. '*n*' polarizing sheets are arranged such that each makes an angle 45° with the preceding sheet. An unpolarized light of intensity I is incident into this arrangement. The output intensity is found to be $\frac{I}{64}$. The value of *n* will be:

(*a*) 3 (*b*) 6 (*c*) 5 (*d*) 4

83. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): If three polarisers are arranged such that the axis of any two successive polarisers make equal angle of 45° with each other. If unpolarised light of intensity I_0 is incident on first polariser then intensity of emergent light after 3rd polariser is $\frac{I_0}{8}$.

Reason (R): In given condition each time intensity becomes 50% by Malus law.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is Not the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true

84. Which of the following phenomena is not common to sound and light waves?

(*a*) Interference (*b*) Diffraction

(*c*) Polarisation (*d*) Reflection

Polarisation by Reflection and Refraction

85. Unpolarised light is incident from air on a plane surface of a material of refractive index 'μ'. At a particular angle of incidence '*i*', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

(*a*) Reflected light is polarised with its electric vector parallel to the plane of incidence.

(*b*) Reflected light is polarised with its electric vector perpendicular to the plane of incidence.

(*c*) $i = \tan^{-1}\left(\frac{I}{\mu}\right)$ (*d*) $i = \sin^{-1}\left(\frac{I}{\mu}\right)$

86. When incident light is polarized in a plane perpendicular to the plane of incidence and it is incident at the angle of polarization.

87. Reflected light is perfectly polarised

88. Refracted light will be partially polarised.

89. Reflected light is partially polarised.

90. There will be no reflection.

(*a*) 1, 2 (*b*) 2, 3 (*c*) 2, 4 (*d*) 4

87. When unpolarised light beam is incident from air glass (n = 1.5) at the polarising angle:

(a) reflected beam is polarised 100%

(b) reflected and refracted beams are partially polarized.

(c) the reason for option (a) is that almost all the light is reflected

(d) All of the above.

88. The critical angle of a certain medium is $\sin^{-1}\left(\frac{3}{5}\right)$. The polarizing angle of the medium is:

(a) $\sin^{-1}\left(\frac{4}{5}\right)$ (b) $\tan^{-1}\left(\frac{5}{3}\right)$

(c) $\tan^{-1}\left(\frac{3}{4}\right)$ (d) $\tan^{-1}\left(\frac{4}{3}\right)$

89. Brewster's angle in terms of refractive index (μ) of the medium.

(a) $\tan^{-1}\sqrt{\mu}$ (b) $\sin^{-1}\mu$

(c) $\sin^{-1}\sqrt{\mu}$ (d) $\tan^{-1}\mu$

90. The Brewster's angle i_b for an interface should be:

(a) $30° < i_b < 45°$ (b) $45° < i_b < 90°$

(c) $i_b = 90°$ (d) $0° < i_b < 30°$

91. If the light is polarised by reflection, then the angle between reflected and refracted light is:

(a) π (b) π/2

(c) 2π (d) π/4

92. In the propagation of light waves, the angle between the plane of vibration and the plane of polarisation is:

(a) 0° (b) 90°

(c) 45° (d) 180°

93. When unpolarized light is incident at an angle of 60° on a transparent medium from air. The reflected ray is completely polarized. The angle of refraction in the medium is:

(a) 30° (b) 60°

(c) 90° (d) 45°

94. When light wave travelling in air suffers reflection at the interface between air and glass, the change of phase of the reflected wave is equal to:

(a) zero (b) π/2

(c) π (d) 2π

95. Choose the correct statement:

(a) the Brewster's angle is independent of wavelength of light.

(b) the Brewster's angle is independent of nature of reflection surface.

(c) the Brewster's angle is different for different wavelengths.

(d) Brewster's angle depends on wavelength but not on the nature of reflection.

Rainbow

96. Pick the wrong answer in the context with rainbow.

(a) Rainbow is a combined effect of dispersion, refraction and reflection of sunlight.

(b) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.

(c) The order of colours is reversed in the secondary rainbow.

(d) An observer can see a rainbow when his front is towards the sun.

97. Rainbow is formed due to:

(a) scattering and refraction

(b) internal reflection and dispersion

(c) reflection only

(d) diffraction and dispersion

Scattering of Light

98. The reddish appearance of the sun at sunrise and sunset is due to:

(a) the scattering of light

(b) the polarization of light

(c) the colour of the sun

(d) the colour of the sky

99. The blue colour of the sky is due to the phenomenon of:

(a) scattering (b) dispersion

(c) reflection (d) refraction

Validity of Ray Optics and Wave Optics

100. For what distance is ray optics a good approximation when a plane light wave is incident on a circular aperture of width 2 mm having wavelength 600 nm?

(a) 6.7 m (b) 3.5 m

(c) 2.7 m (d) 5.7 m

Answer Key

1. Light travels from the sun to earth through the vacuum of space because light waves do not require a medium to pass through. Light is an electromagnetic wave, and electromagnetic waves do not require a medium to propagate.

2. (*a*)	**3.** (*b*)	**4.** (*b*)	**5.** (*b*)	**6.** (*c*)	**7.** (*d*)	**8.** (*a*)	**9.** (*d*)	**10.** (*d*)	
11. (*b*)	**12.** (*a*)	**13.** (*a*)	**14.** (*a*)	**15.** (*c*)	**16.** (*a*)	**17.** (*b*)	**18.** (*b*)	**19.** (*b*)	**20.** (*b*)
21. (*d*)	**22.** (*a*)	**23.** (*b*)	**24.** 8 maxima, 8 minima		**25.** 5 sin ($kx + \omega t + 67°$)		**26.** 3 sin ($kx + \omega t + 60°$)		**27.** $5\sqrt{3}$
28. (4)	**29.** 5.1×10^{-4} m		**30.** ($4I_0$)	**31.** (*d*)	**32.** 8	**33.** (*c*)	**34.** (*a*)	**35.** (*b*)	**36.** (*d*)
37. (*d*)	**38.** (*c*)	**39.** (*d*)	**40.** (*b*)	**41.** (*d*)	**42.** (*c*)	**43.** (*b*)	**44.** (*c*)	**45.** (*d*)	**46.** (*a*)
47. (*c*)	**48.** Red light		**49.** $I_0/4$	**50.** (*a*)	**51.** 5×10^{-6}	**52.** 6Å	**53.** (*c*)	**54.** (*a*)	**55.** (9)
56. (3)	**57.** (*b*)	**58.** (*b*)	**59.** (*a*)	**60.** (*a*)	**61.** (0.2)	**62.** (*b*)	**63.** (*d*)	**64.** (*b*)	**65.** (*c*)
66. (*b*)	**67.** (*c*)	**68.** (*d*)	**69.** (*d*)	**70.** (*b*)	**71.** (*c*)	**72.** (*d*)	**73.** (*c*)	**74.** (*b*)	**75.** 198
76. (*b*)	**77.** (*c*)	**78.** (*c*)	**79.** (*c*)	**80.** (*d*)	**81.** (*b*)	**82.** (*b*)	**83.** (*a*)	**84.** (*c*)	**85.** (*b*)
86. (*d*)	**87.** (*a*)	**88.** (*b*)	**89.** (*d*)	**90.** (*b*)	**91.** (*b*)	**92.** (*b*)	**93.** (*a*)	**94.** (*c*)	
95. (*c*)	**96.** (*d*)	**97.** (*b*)	**98.** (*a*)	**99.** (*a*)	**100.** (*a*)				